Estimating the Probability of Winning a VALORANT Match Using Historical Player Performance

1. Motivation

VALORANT is a competitive team-based first-person shooter in which individual performance does not necessarily guarantee a team victory. A player may perform well individually but still lose because match outcomes can also depend on teammates, opponents, team coordination, and other factors. This raises an interesting question: can a player's past performance help estimate their probability of winning the next match?

This project explores whether historical player statistics can provide useful information for estimating the probability of winning a subsequent VALORANT competitive match. Rather than attempting to determine whether a player will definitely win or lose, the project focuses on producing a probability estimate based on information available before the match.

The project is relevant to predictive modeling because it distinguishes between predicting a definite outcome and estimating the likelihood of that outcome. It also examines the limitations of using individual statistics to predict results in a team-based game.

2. Problem Definition

2.1 Goal, Features, Target, and Loss

The goal is to estimate the probability that a player’s team wins a target competitive match using the player’s performance in previous matches.

The research question is:

Can a player's historical performance statistics be used to estimate the probability of winning their next VALORANT competitive match?

The input features will be calculated from the player's previous five matches and may include average Average Combat Score (ACS), kill/death ratio (K/D), KAST percentage, headshot percentage, damage differential per round, and historical win rate over previous five matches.

This is a supervised binary classification problem. The target variable (Y) represents the outcome of a target match, where 1 indicates a win and 0 indicates a loss. The model will

estimate the probability that the player wins the target match using only information from matches played before it.

The target match's own performance statistics will not be used as predictors because they are not available before the match outcome occurs.

Binary cross-entropy, commonly called log loss in binary classification, will be the primary evaluation metric because the goal is to estimate winning probabilities. Accuracy will be used as a secondary metric to measure the proportion of correctly classified outcomes.

2.2 Potential Impact

This project may help players understand whether their recent performance statistics provide useful information about future match outcomes. It can also demonstrate the limitations of individual-statistic-based prediction in a team-based game.

3. Related Work

Previous research on esports prediction provides a foundation for this project. Bahrololloomi et al. (2023) investigated player performance metrics for predicting League of Legends match outcomes while considering player roles. Their work provides an example of how individual performance statistics can be used to study competitive outcomes. However, results from League of Legends may not directly generalize to VALORANT because the games have different mechanics and team dynamics.

Birant and Birant (2023) studied multi-objective, multi-instance learning for esports. Their work highlights the complexity of modeling esports outcomes using data involving multiple players and objectives. This is relevant because an individual player's statistics may not capture all factors that influence a team's performance.

Hayakawa et al. (2025) explored VALORANT round-outcome prediction using tactical features derived from video analysis. Their work focuses on tactical information at the round level, whereas this project focuses on historical player performance statistics to estimate the probability of winning a subsequent match.

Together, these studies demonstrate different approaches to esports prediction. This project takes a more focused approach by examining whether historical player-level statistics can help estimate the probability of winning the next VALORANT match.

4. Data

4.1 Data Source and Collection

The data for this project were collected from VALORANT Tracker, a third-party platform that provides player performance statistics and match histories. The dataset includes 40 players, their performance statistics of the current competitive season, and their 10 most recent recorded matches. The data represent a limited sample collected for this course project rather than the entire VALORANT player population.

4.2 Dataset Description

The dataset contains two tables:

- **Player-level data:** 40 records containing overall statistics, including win rate, K/D, ACS, KAST percentage, damage differential per round, and headshot percentage.
- **Player-match data:** 400 player-match observations containing match outcomes and performance statistics, including ACS, K/D, KAST percentage, headshot percentage, and damage differential per round.

Across the 400 player-match observations, 208 are wins and 192 are losses, corresponding to an observed win proportion of 52% and a loss proportion of 48%.

Match 1 represents the most recent match, while Match 10 represents the oldest match. A fixed five-match historical window will be used to estimate the outcomes of Matches 1 through 5. For example, Match 1 will be estimated using Matches 2 through 6, while Match 5 will be estimated using Matches 6 through 10. This procedure can produce up to five target observations per player, or 200 player-target observations in total.

4.3 Data Limitations

The dataset has several limitations. First, it contains only 40 players and may not represent the broader VALORANT player population. Second, it does not include potentially important contextual information such as maps, agents, opponents, team compositions, or verified match dates. Third, the match identifiers are player-specific indices and do not establish whether records from different players belong to the same actual match.

Because VALORANT is a team-based game, individual historical statistics may not capture all factors affecting match outcomes. Therefore, the estimated probabilities should be interpreted as predictions based on the available features, not as exact probabilities that apply to every player or match.

The player-level overall statistics of the current season will mainly be used for descriptive purposes because they may include information outside the historical window for a particular prediction. Predictive features will instead be constructed from the five matches preceding each target match.

5. Analysis Plan

5.1 Models

We plan to use logistic regression as the primary model because it estimates the probability of a binary outcome and is relatively easy to interpret. We will also consider a decision tree as a comparison model. A decision tree can capture nonlinear relationships between performance statistics and match outcomes, but it may overfit the small dataset.

5.2 Training the Models

We will first clean the data and use each player's previous five Competitive matches to calculate historical performance features, including ACS, K/D, KAST%, HS%, DDΔ/Round, and win rate. The target will be the outcome of the next recorded match, encoded as 1 for a win and 0 for a loss. The target match's own performance statistics will not be used as input features.

We will split the data into training and test sets by player ID so that the same player does not appear in both sets. The models will be trained on the training set and evaluated on the test set. Log loss will be the primary evaluation metric because our project focuses on predicting win probabilities. Accuracy and ROC-AUC may also be used as secondary metrics.

5.3 Baselines

We will compare the models with two baselines.

The first will predict the overall win rate in the training set for every test observation. This provides a simple reference for evaluating whether our models improve on a constant probability prediction.

The second baseline will use logistic regression with only the player's win rate over the previous five Competitive matches as the input feature. This allows us to compare the predictive value of recent win rate alone with that of the full model, which also uses ACS, K/D, KAST%, HS%, and DDΔ/Round.

References

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